Practical 1

Electricity and Electronics EBN 111

12-16 April 2021

Practical Information			
Maximum marks:	50	Full marks:	50
Duration of practical:	180 min	Open/Closed book:	Open
Total number of pages (this page included)		8	

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Important

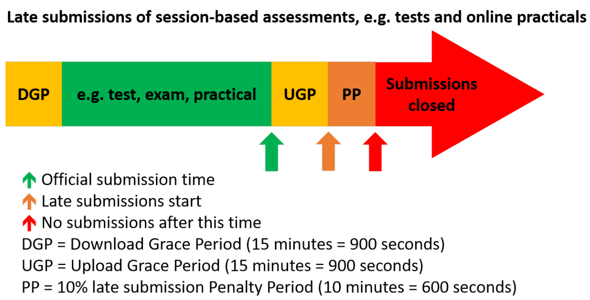
1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write irrational numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your practical, you declare the following:**

   The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all practicals for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the practical for students to download all documents related to the practical and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the practical.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the practical.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.practicals@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final practical mark.
8. No late submissions is accepted after the PP.

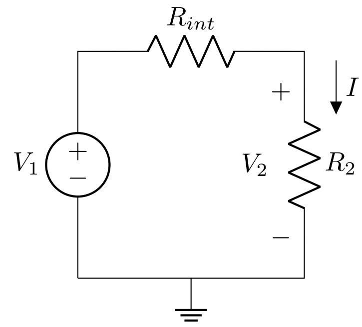


Section 1

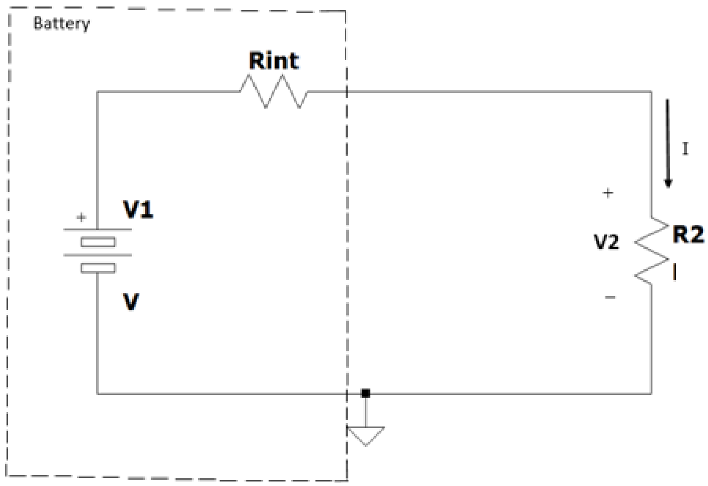
Questions 1 to 2 [4]

Draw the circuit schematic of Circuit A in LTspice as shown in Circuit B with $V_1 = [V1]$, $R_{int} = [Rint]$, and $R_2 = [R2]$. Simulate the circuit and measure the voltage V_2 and current I associated with resistor R_2.

Circuit A - Schematic:



Circuit B - LTSpice:

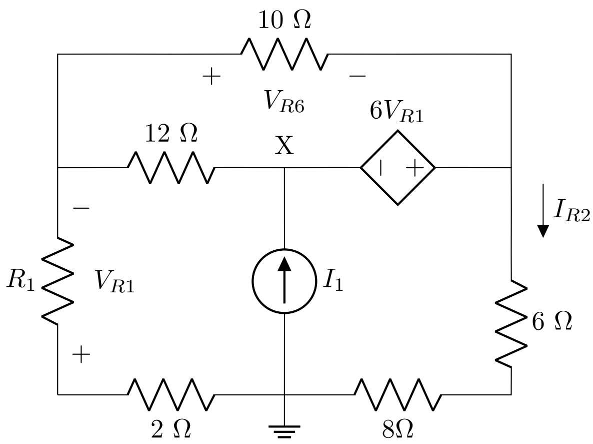


01	V_2 [2]
02	I [2]

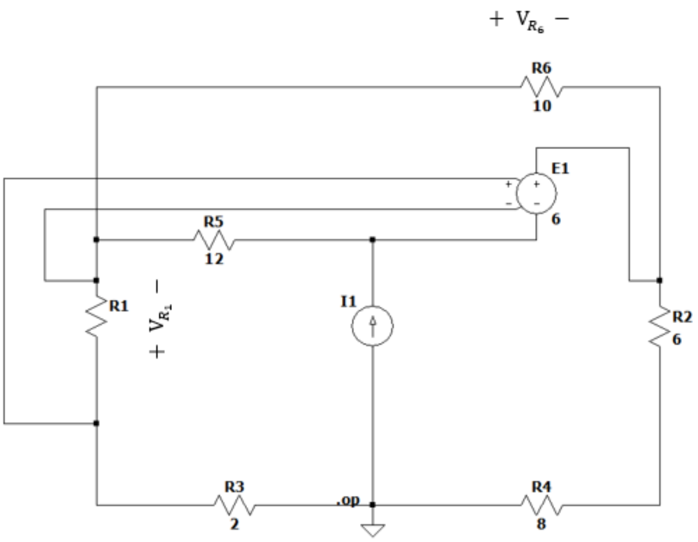
Questions 3 to 6 [8]

Draw the circuit schematic of Circuit A in LTspice as shown in Circuit B with R_1 = [$R1$], and I_1 = [$I1$]. Simulate the circuit and measure V_{R1}, V_{R6}, I_{R2}, and V_X at node X.

Circuit A - Schematic:

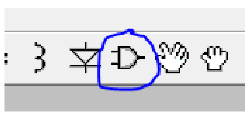


Circuit B - LTSpice:

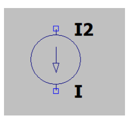


Hints:

- The DC current source part name is *current*. To obtain it, click on the *component* tool available on the tool bar as illustrated below:



In the select component symbol dialogue box that pops up, double click on *current* and use Ctrl + R to rotate (if need be). The DC current source component is shown below:

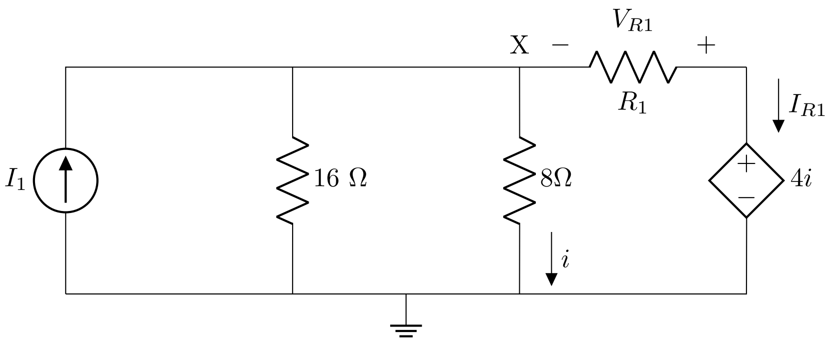


	• The voltage-controlled voltage source part name is *e*. To obtain it, click on the *component* tool available on the tool bar. Double click on the *e* available in the second column just under the component named *diode*. Pay particular attention to the polarity of this element. • Remember to always ground your circuit to ensure a complete path for current flow.
03	V_{R1} [2]
04	V_{R6} [2]
05	I_{R2} [2]
06	V_X [2]

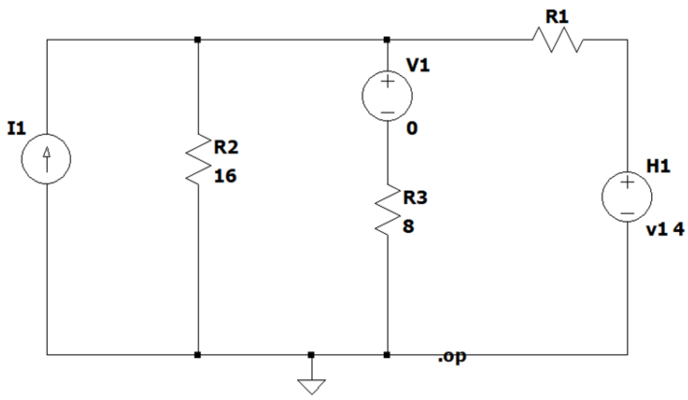
Questions 7 to 11 [10]

Draw the circuit schematic of Circuit A in LTspice as shown in Circuit B with $R_1 = [R1]$, and $I_1 = [I1]$. Simulate the circuit and measure V_X at node X, I_{R1}, and V_{R1}. Also measure the the total power absorbed $P_{absorbed}$, and the total power supplied $P_{supplied}$ (include the appropriate sign where applicable).

Circuit A - Schematic:



Circuit B - LTSpice:



Hints:

- As shown in Circuit B, $V1$ is a voltage source that you will insert in a series connection with $R3$ for the purpose of measuring current. The value 4 is the gain of the current dependent voltage source. You can add the voltage source $V1$ by selecting the *component* tool and searching for *voltage* among the list of components that pop up. Set the value of the voltage source to zero.
- The current-controlled voltage source can be found by using the *component* tool and searching for the linear current dependent voltage source using the letter h. Double click on the letter and place it on the workspace. Right click on the H symbol that you have placed and change the value from H to "V1 4". Note that there must be a space between $V1$ and 4.
- To calculate the total power absorbed and supplied in the circuit, please see the Practical 1 Tutorial for more information, specifically in the "**Tip**" provided under Fig. 24. You can also consider the voltage and current for individual components in the circuit to obtain the power in each individual component. The power absorbed should balance the power supplied to the circuit.

07	V_X [2]
08	I_{R1} [2]
09	V_{R1} [2]
10	$P_{absorbed}$ [2]
11	$P_{supplied}$ [2]

Questions 12 to 25 [28]

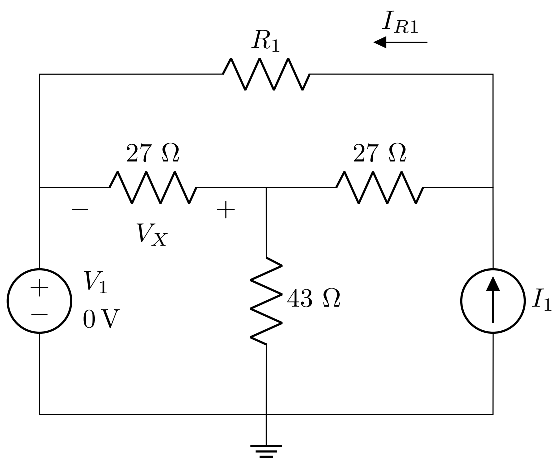
Draw the circuit schematic of Circuit A in LTspice as shown in Circuit B with R_1 = [$R1$], and I_1 = [$I1$].

Using DC Sweep Analysis, sweep the voltage source $V1$ over the range of 0 to 100 V in increments or steps of 5 V and measure the current I_{R1} and the voltage V_X. Using the graphs, measure the minimum and maximum values of the current I_{R1} and the voltage V_X.

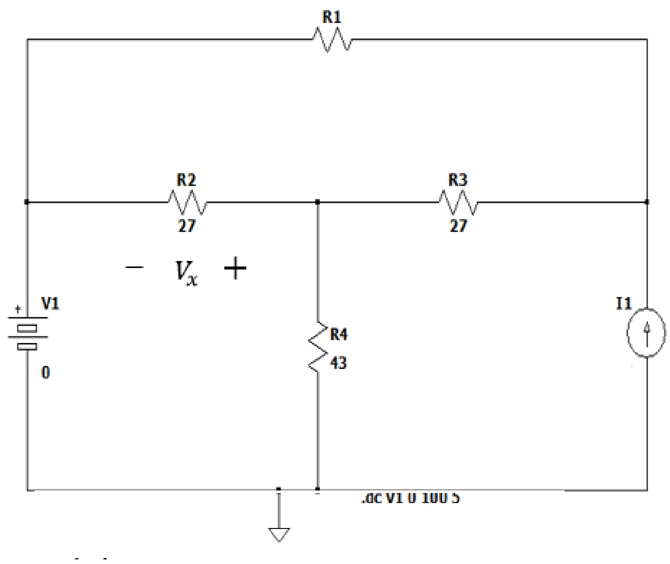
For questions 16 to 20, use the voltage measurement graph of V_X to find the value of V_X at the given V_1 values.

For questions 21 to 25, set the voltage source to V_1 = 0 V. Perform a DC Sweep Analysis again, but this time use the current source I_1 as the sweep variable. Sweep I_1 from 0 A to 5 A in 200 mA increments and measure the current I_{R1} through $R1$. Use the current measurement graph of I_{R1} to find the value of I_{R1} at the given I_1 values.

Circuit A - Schematic:



Circuit B - LTSpice:



Hints:

- To place the voltage source as shown in Circuit B, click on the *component* tool and search for *Misc* which can be found in the first column of component list. Double click on it and choose *battery*. Place your battery on the workspace and assign the appropriate voltage value to it. Note that either a voltage source or battery can be used.
- After running the simulation, right-click the plot layout and add the required trace to your graph. To take readings from your curve, click on the label at the top (middle) of your graph which indicates the quantity and component you are measuring from. Move the cursor to read accurately from the correct positions as required to answer the questions below. For ease of reading values from the graphs, try to have only one trace on your plot at a time.
- Remember to always ground your circuit to ensure a complete path for current flow.

12	I_{R1min} [2]
13	I_{R1max} [2]
14	V_{Xmin} [2]
15	V_{Xmax} [2]
16	V_X if $V_1 = [V1a]$ [2]
17	V_X if $V_1 = [V1b]$ [2]
18	V_X if $V_1 = [V1c]$ [2]
19	V_X if $V_1 = [V1d]$ [2]
20	V_X if $V_1 = [V1e]$ [2]
21	I_{R1} if $I_1 = [I1a]$ [2]
22	I_{R1} if $I_1 = [I1b]$ [2]
23	I_{R1} if $I_1 = [I1c]$ [2]
24	I_{R1} if $I_1 = [I1d]$ [2]
25	I_{R1} if $I_1 = [I1e]$ [2]

Grand Total : [50]